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Semiconductor structure and method of manufacturing the same

Abstract

A semiconductor structure and its manufacturing method are provided. A semiconductor structure includes a substrate and several bit lines on the substrate. Each of the bit lines includes a first conductive layer on the substrate, a second conductive layer on the first conductive layer, and a hardmask layer on the second conductive layer. The semiconductor structure further includes several contacts disposed on the substrate and positioned between two adjacent bit lines, wherein the bottom surfaces of the contacts physically contact the substrate. The top surfaces of the contacts are not higher than the top surfaces of the hardmask layers. Each of the contacts includes a bottom contact part on the substrate and a top contact part on the bottom contact part, and a width of a top surface of the top contact part is greater than a width of a top surface of the bottom contact part.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Taiwan Application No. 110100933, filed on Jan. 11, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

Field of the Disclosure

(2) The present disclosure relates to a semiconductor structure and its manufacturing method, and in particular, it relates to a semiconductor structure of dynamic random access memory (DRAM) and its manufacturing method.

Description of the Related Art

(3) As semiconductor technology improves, the memory units used in dynamic random access memory (DRAM) devices are increasingly miniaturized, while the degree of integration of the memory units is increased to comply with consumer demand for miniaturized electronic devices. The development of the buried word line dynamic random access memory device was designed to

satisfy the need to increase the degree of integration of dynamic random access memory devices, in order to accelerate the operating speed of the device. Furthermore, isolation structures (such as shallow trench isolation structures) may be developed further within the buried word line dynamic random access memory device, to prevent interference between word lines.

(4) In conventional processes for forming dynamic random access memory devices, doping regions are formed within the active regions in the substrate, and the doping regions act as source regions and drain regions at two opposite sides of the bit lines. The memory device further includes contacts that electrically connect the surfaces of the source regions and drain regions. In a conventional dynamic random access memory device, pillar-like structures that include the contacts extending over the top surfaces of the bit lines and the contact plugs landing on the contacts are formed. Since the integration of memory devices continues to increase, the space between related components (such as bit lines, contacts and contact plugs) is also constantly shrinking. More accurate manufacturing processes (such as immersion lithography) are required to complete the semiconductor structure of the existing memory device. However, this also greatly increases the production cost.

SUMMARY

(5) A semiconductor structure is provided. The semiconductor structure includes a substrate and several bit lines on the substrate. In some embodiments, each of the bit lines includes a first conductive layer on the substrate, a second conductive layer on the first conductive layer, and a hardmask layer on the second conductive layer. In some embodiments, the semiconductor structure also includes contacts disposed on the substrate and positioned between two adjacent bit lines. In some embodiments, the bottom surfaces of the contacts physically contact the substrate, and the top surfaces of the contacts are not higher than the top surfaces of the hardmask layers adjacent to the contacts. Also, each of the contacts includes a bottom contact part on the substrate and a top contact part on the bottom contact part, and a width of a top surface of the top contact part is greater than a width of a top surface of the bottom contact part.

(6) A method of manufacturing a semiconductor structure is provided. The method includes providing a substrate and bit lines formed on the substrate. Also, a spacer material layer covers the sidewalls of the top surface of each of the bit lines. In some embodiments, each of the bit lines includes a conductive layer and a hardmask layer on the conductive layer. In some embodiments, the method also includes forming a first contact material layer on the substrate, wherein the first contact material layer covers the spacer material layers on the bit lines. In some embodiments, the method also includes etching the first contact material layer to remove a portion of the first contact material layer, a portion of the hardmask layers and a portion of the spacer material layers, thereby recessing the first contact material layer and forming the first recesses between the remaining portions of the hardmask layers, wherein the remaining portions of the first contact material layer form the bottom contact parts. In some embodiments, the method further includes depositing the first dielectric layer over the bottom contact parts, wherein the first dielectric layer is conformally formed on the sidewalls and the bottom surfaces of the first recesses and the top surfaces of the remaining portions of the hardmask layers. In some embodiments, the method also includes patterning the first dielectric layer to remove a portion of the first dielectric layer and expose the top surfaces of the bottom contact parts, thereby forming spacers with different thicknesses on two opposite sidewalls of the hardmask layers that protrude from the bottom contact parts. The second recesses are formed between two adjacent spacers, and the second recesses expose the top surfaces of the bottom contact parts. In some embodiments, the method further includes forming the top contact parts on the bottom contact parts, wherein the top surfaces of the top contact parts are not higher than the top surfaces of the hardmask layers adjacent to the top contact parts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-4**, **5A-5C**, **6A**, **6B**, **7A-7C**, **8A**, **8B**, **9A**, **9B**, **10A**, **10B**, **11A**, **11B**, **12A**, **12B**, **13A**, **13B**, **14A**, **14B**, **15A-15C**, **16A-16C** illustrate the cross-sectional views and the top views of intermediate stages in a method of manufacturing a semiconductor structure in an electronic, in accordance with some embodiments of the present disclosure, wherein FIGS. **5C**, **7C**, **15C** and **16C** are top views of substrates of intermediate stages of a method of manufacturing the semiconductor structure, in accordance with some embodiments;

(2) FIGS. **5A**, **7A**, **15A** and **16A** are cross-sectional views taken along sectional lines A-A' of the structures of FIGS. **5C**, **7C**, **15C** and **16C**, respectively; and

(3) FIGS. **5B**, **7B**, **15B** and **16B** are cross-sectional views taken along sectional lines B-B' of the structures of FIGS. **5C**, **7C**, **15C** and **16C**, respectively.

(4) FIG. **17** is a cross-sectional view of a semiconductor structure, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

(5) The present disclosure provides a semiconductor structure and a method of manufacturing the same, wherein the semiconductor structure includes contacts not protruding from the top surfaces of the bit lines. In the conventional method, the pillar-like structures that include the contacts extending over the top surfaces of the bit lines and the contact plugs landing on the contacts are formed. However, it requires complicated and expansive processes to form those pillar-like structures. Compared with the complicated and expansive conventional method for forming pillar-like structures, the semiconductor structure and the method of manufacturing the same as provided in accordance with some embodiments decrease the process complexity and save the production cost. Especially when the integration of memory devices continues to increase, the spacing between related components (such as bit lines, contacts and contact plugs) is also constantly shrinking. The manufacturing method provided in the embodiment does not require expensive manufacturing processes (such as immersion lithography) to complete the fabrication of semiconductor structures. According to some embodiments, the semiconductor structure and the method of manufacturing the same not only greatly decrease the manufacturing process cost, but also obtain the structure having components with complete profiles. Therefore, the semiconductor structure manufactured by the method of the embodiment has great electrical characteristics.

(6) In addition, the semiconductor structure of the embodiment can be, for example, applied to a dynamic random access memory (DRAM) device for manufacturing the bit lines and the contacts of the memory device. For the sake of simplicity and clarity, three bit lines and two contacts on the substrate are illustrated in the drawings for describing the semiconductor structure in some embodiments of the present disclosure. However, the disclosure is not limited in those illustrating structures. In addition, active regions in the substrate can be defined by several isolation structures. Also, other components, such as buried word lines (i.e. the gate electrodes), are further formed in the substrate. For the sake of simplicity and clarity, those elements formed in the substrate **100** are omitted in the drawings.

(7) FIGS. **1-4**, **5A-5C**, **6A**, **6B**, **7A-7C**, **8A**, **8B**, **9A**, **9B**, **10A**, **10B**, **11A**, **11B**, **12A**, **12B**, **13A**, **13B**, **14A**, **14B**, **15A-15C**, **16A-16C** illustrate the cross-sectional views and the top views of intermediate stages in a method of manufacturing a semiconductor structure in an electronic, in accordance with some embodiments of the present disclosure. FIGS. **5C**, **7C**, **15C** and **16C** are top views of substrates of intermediate stages of a method of manufacturing the semiconductor structure, in accordance with some embodiments. FIGS. **5A**, **7A**, **15A** and **16A** are cross-sectional views taken along sectional lines A-A' of the structures of FIGS. **5C**, **7C**, **15C** and **16C**, respectively. FIGS. **5B**, **7B**, **15B** and **16B** are cross-sectional views taken along sectional lines B-B' of the structures of FIGS. **5C**, **7C**, **15C** and **16C**, respectively.

(8) Referring to FIG. **1**, a substrate **100** is provided, and several bit lines **116** are formed on the

substrate **100**. Also, examples of the elements formed in the substrate **100** include several isolation regions extending downwards from the surface of the substrate **100** to define active regions, several buried word lines separated from the bit lines **116**, and an insulating layer in the substrate **100** for separating the bit lines **116** and word lines. For the sake of simplicity and clarity, those elements formed in the substrate **100** are omitted in the drawings. In some embodiments, the substrate **100** includes one or more semiconductor materials. For example, the substrate **100** may include silicon, gallium arsenide, gallium nitride, germanium silicide, another suitable substrate material, or a combination thereof. In some embodiments, the substrate **100** is a silicon-on-insulator (SOI) substrate.

(9) As shown in FIG. 1, each of the bit lines **116** includes the first conductive layer **113** on the top surface **100a** of the substrate **100**, the second conductive layer **114** on the first conductive layer **113** and the hard mask (HM) layer **115** on the second conductive layer **114**. The first conductive layer **113** includes an epitaxial material or polysilicon. The second conductive layer **114** may include a metal material (such as tungsten), and the hard mask layer **115** may include a nitride material (such as silicon nitride). However, the disclosure is not limited to the materials provided herein. In addition, those bit lines **116** are spaced apart from each other in the first direction D1 and extend in the second direction D2. The first conductive layer **113**, the second conductive layer **114** and the hard mask layer **115** are stacked in the third direction D3. Also, the buried word lines (not shown) in the substrate **100** extend in the first direction D1.

(10) In addition, spacer material layers cover the sidewalls of the bit lines **116** to protect the bit lines **116**. As shown in FIG. 1, a first nitride material layer **12-1** is formed on the sidewalls of each of the bit lines **116**, an oxide material layer **12-2** is formed on the outer sidewalls of the first nitride material layer **12-1**, and a second nitride material layer **12-3** is formed on the oxide material layer **12-2**. The second nitride material layer **12-3** is deposited on the outer sidewalls of the oxide material layer **12-2** and also covers the top surfaces of the first nitride material layer **12-1** and the hard mask layer **115**. In each of the bit lines **116**, the first nitride material layer **12-1** extends on the sidewalls of the hard mask layer **115**, the second conductive layer **114** and the first conductive layer **113**. Also, the first nitride material layer **12-1** and the second nitride material layer **12-3** may include (but not limited to) silicon nitride, and the oxide material layer **12-2** may include (but not limited to) silicon oxide.

(11) As shown in FIG. 1, a first contact material layer **1250** is formed on the substrate **100** and covers the spacer material layers on the bit lines **116**, wherein the first contact material layer **1250** fills the space between adjacent bit lines **116**. In some embodiments, the first contact material layer **1250** covers the top surfaces and sidewalls of the spacer material layers around all of the bit lines **116**, and also covers the exposed surface of the substrate **100** between the bit lines **116**. As shown in FIG. 1, the top surface of the first contact material layer **1250** is higher than the top surfaces of the bit lines **116**. The first contact material layer **1250** may include doped or undoped polysilicon, and can be formed on the substrate **100** by chemical vapor deposition. In some embodiments, after the patterning processes are performed subsequently, the first contact material layer **1250** will become a part of the contact (hereinafter referred as the bottom contact part).

(12) Next, referring to FIG. 2, a portion of the first contact material layer **1250** is removed to recess the first contact material layer **1250**. The remaining portions of the first contact material layer **1250** form the bottom contact parts **125**. In some embodiments, the portion of the first contact material layer **1250** can be removed by an etch back process. Also, in this etch back process, a portion of the hard mask layer **115** and portions of the spacer material layers (including a portion of the second nitride material layer **12-3**, a portion of the oxide material layer **12-2** and a portion of the first nitride material layer **12-1**). Therefore, after this etch back step is performed, the top surfaces of the remaining portions of the spacer material layers are substantially coplanar. Also, the top surfaces **125a** of the bottom contact parts **125** are lower than the top surfaces of the remaining portions of the spacer material layers.

(13) Next, referring to FIG. 3, the portions of the spacer material layers on the sidewalls of the hardmask layers **115** that protrude from the bottom contact parts **125** are removed. According to some embodiments, the portions of the second nitride material layer **12-3** and the oxide material layer **12-2** on the sidewalls of the hardmask layers **115** that protrude from the bottom contact parts **125** can be removed by a suitable wet etching process. Also, in some embodiments, the wet etching process may slightly damage the hard mask layers **115** and the bottom contact parts **125**, which leads to material loss. Therefore, the heights of the bottom contact parts **125** and the hard mask layers **115** as shown in FIG. 3 would be slightly lower than the heights of the bottom contact parts **125** and the hard mask layers **115** as shown in FIG. 2.

(14) As shown in FIG. 3, after the bottom contact parts **125** are formed followed by removing portions of the spacer material layers, the remaining portions of the hard mask layer **115** protrude from the top surfaces **125a** of the bottom contact parts **125**. The first nitride layers **121** as formed cover the sidewalls of the hard mask layers **115**, the sidewalls of the second conductive layer **114** and the sidewalls of the first conductive layer **113**. Also, the top surfaces of the oxide layers **122** and the top surfaces of the second nitride layers **123** are substantially level with the top surfaces **125a** of the bottom contact parts **125**. According to some embodiments, after the bottom contact parts **125** are formed followed by removing portions of the spacer material layers, the first recesses **131** are formed between remaining portions of the hard mask layers **115**. Accordingly, the first recesses **131** expose the top surfaces **125a** of the bottom contact parts **125**, the top surfaces **122a** of the oxide layers **122** and the top surfaces **123a** of the second nitride layers **123**.

(15) In addition, as shown in FIG. 3, in each of the bit lines **116**, the portions of the hard mask layer **115** that protrude from the top surfaces **125a** of the bottom contact parts **125** have the first sidewalls **115S1** and the second sidewalls **115S2** opposite to the first sidewalls **115S1**. In this stage, the first nitride layers **121** having the same thickness are formed on the first sidewalls **115S1** and the second sidewalls **115S2** of the hard mask layers **115**. In the subsequent processes, the spacers having different thicknesses are formed on the opposite sidewalls of the hard mask layers **115** that protrude from the top surfaces **125a** of the bottom contact parts **125**. Therefore, two opposite sidewalls of the top contact parts formed subsequently contact the spacers having different thicknesses. Thus, according to the embodiments of the disclosure, the positions of the top contact parts of the adjacent rows are staggered by changing and adjusting the thickness of the spacers, so that the positions of the contact plugs that are connected to the underlying top contact parts can be misaligned with each other.

(16) Next, referring to FIG. 4, a first dielectric layer **132** is conformally deposited over the bottom contact parts **125** to cover the sidewalls and bottom surfaces of the first recesses **131**. That is, the first dielectric layer **132** is conformally formed to cover the top surfaces **115a**, the first sidewalls **115S1** and the second sidewalls **115S2** of the hardmask layers **115**. Also, the first dielectric layer **132** further covers the top surface **122a** of the oxide layers **122**, the top surfaces **123** of the second nitride layers **123** and the top surfaces **125a** of the bottom contact parts exposed by the first recesses **131**. In some embodiments, the first dielectric layer **132** has a sufficient thickness to cover at least the top surfaces **122a** of the oxide layers **122** and the top surfaces **123a** of the second nitride layers **123**. In some embodiments, the thickness of the first dielectric layer **132** is (but not limited to) within a range of about 5 nm to about 7 nm.

(17) In addition, the first dielectric layer **132** is a nitride-containing layer, such as a silicon nitride layer. However, the present disclosure is not limited to the material provided herein. The first dielectric layer **132** may include one or more dielectric materials. In some embodiments, the first dielectric layer **132** and the first nitride layer **121** include the same material. The first dielectric layer **132** can be formed by deposition, such as chemical vapor deposition or another suitable deposition method.

(18) Next, a patterning process is performed on the first dielectric layer **132**, so that the spacers with different thicknesses can be formed on two opposite sidewalls of the hardmask layers that

protrude from the bottom contact parts. Also, for the spacers on the sidewalls of the hardmask layers that are adjacent to the first recesses **131** positioned in adjacent rows, the thickness arrangements of the spacers are different, so that the positions of the subsequently formed top contact parts in adjacent rows can be misaligned with each other. FIG. 5A to FIG. 14B illustrate one of the methods of patterning the first dielectric layer **132**, in accordance with some embodiments of the present disclosure.

(19) Referring to FIGS. 5A, 5B and 5C. After the first dielectric layer **132** is formed (FIG. 4), an etching barrier layer **134** is conformally deposited on the first dielectric layer **132**, and a patterned mask layer (such as a patterned photoresist layer) is provided on the etching barrier layer **134** to cover parts of the first recess **131** (FIG. 3). In some embodiments, the etching barrier layer **134** is an undoped polysilicon layer, or another suitable barrier material.

(20) As shown in FIG. 5C, the first recesses **131** are arranged along the first direction **D1** to form several rows, and those rows are spaced apart from each other in the second direction **D2**. For the sake of simplicity and clarity, four rows of the first recess **131** are depicted in FIG. 5C for illustration. The first recesses **131** are arranged as the first recesses **131** of the i -th row $R_{sub.iT}$, the first recesses **131** of the $(i-1)$ -th row $R_{sub.(i-1)T}$, the first recesses **131** of the $(i-2)$ -th row $R_{sub.(i-2)T}$ and the first recesses **131** of the $(i-3)$ -th row $R_{sub.(i-3)T}$, wherein i is a positive integer.

(21) As shown in FIG. 5B and FIG. 5C, in some embodiments, the first photoresist pattern **141** is provided on the first recesses **131** of the i -th row $R_{sub.iT}$ and the first recesses **131** of the $(i-2)$ -th row $R_{sub.(i-2)T}$. The first photoresist pattern **141** exposes the etching barrier layer **134** in the first recesses **131** of the $(i-1)$ -th row $R_{sub.(i-1)T}$ and the first recesses **131** of the $(i-3)$ -th row $R_{sub.(i-3)T}$, as shown in FIG. 5A and FIG. 5C.

(22) In some embodiments, the first photoresist pattern **141** includes several photoresist strips, such as the first photoresist strip **1411** covering the etching barrier layer **134** corresponding to the first recesses **131** of the i -th row $R_{sub.iT}$ and another first photoresist strip **1412** covering the etching barrier layer **134** corresponding to the first recesses **131** of the $(i-2)$ -th row $R_{sub.(i-2)T}$. It should be noted that the width $W_{sub.41}$ of the first photoresist strip **1411/1412** in the second direction **D2** is greater than the width $W_{sub.1}$ of the first recess **131** in the second direction **D2**. For example, the width $W_{sub.141}$ of the first photoresist strip **1411/1412** may be equal to the sum of the width $W_{sub.131}$ of the first recess **131** and part of the distance between the first recesses in adjacent rows, as shown in FIG. 5C. Therefore, the subsequent lithography processes can be performed by using a typical lithography and the first photoresist pattern **141** having the first photoresist strips **1411** and **1412** with greater widths provided herein. Thus, whether using immersion lithography (a photolithography resolution enhancement technique) or a general photoresist with a typical photolithography process, the subsequent processes can be completed using the manufacturing method of the embodiment and the first photoresist pattern **141**.

(23) Next, referring to FIG. 6A and FIG. 6B, a first implant step **151** is performed on the portions of the etching barrier layer **134** on the first recesses **131** of the $(i-1)$ -th row $R_{sub.(i-1)T}$ and the $(i-3)$ -th row $R_{sub.(i-3)T}$ that are exposed by the first photoresist pattern **141**. According to some embodiments, the first implant step **151** is performed on the exposed portions of the etching barrier layer **134** at an angle to introduce the dopants of a suitable conductivity type, such as P-type dopants (e.g. boron or another suitable doping material). As shown in FIG. 6A, the positions of the implanted portions of the exposed etching barrier layer **134** correspond to the second sidewalls **115S2** and the top surfaces **115a** of the hard mask layer **115**. As shown in FIG. 6B, the portions of the etching barrier layer **134** that are covered by the first photoresist pattern **141** are not implanted by any dopant.

(24) After the first implant step **151** is completed, the first photoresist pattern **141** is removed. In some embodiments, the first photoresist pattern **141** is removed by an ashing process.

(25) Next, referring to FIGS. 7A, 7B and 7C, in some embodiments, the second photoresist pattern **142** is provided on the first recesses **131** of the $(i-1)$ -th row $R_{sub.(i-1)T}$ and the first recesses **131**

of the (i-3)-th row R.sub.(i-3)T. The second photoresist pattern **142** exposes the etching barrier layer **134** in the first recesses **131** of the i-th row R.sub.iT and the first recesses **131** of the (i-2)-th row R.sub.(i-2)T, as shown in FIG. 7B and FIG. 7C.

(26) In some embodiments, the second photoresist pattern **142** includes several photoresist strips, such as the second photoresist strip **1421** covering the etching barrier layer **134** corresponding to the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and another second photoresist strip **1422** covering the etching barrier layer **134** corresponding to the first recesses **131** of the (i-3)-th row R.sub.(i-3)T. As shown in FIG. 7C, according to some embodiments, the width W.sub.142 of the second photoresist strip **1421/1422** in the second direction D2 is greater than the width W_{in} of the first recess **131** in the second direction D2. For example, the width W.sub.142 of the second photoresist strip **1421/1422** may be equal to the sum of the width W.sub.131 of the first recess **131** and part of the distance between two first recesses in adjacent rows, as shown in FIG. 7C.

Therefore, the subsequent lithography processes can be performed by using a typical lithography and the second photoresist pattern **142** having the second photoresist strips **1421** and **1422** with greater widths provided herein. Thus, whether using immersion lithography (a photolithography resolution enhancement technique) or a general photoresist with a typical photolithography process, the subsequent processes can be completed using the manufacturing method of the embodiment and the second photoresist pattern **142**.

(27) Next, referring to FIG. 8A and FIG. 8B, a second implant step **152** is performed on the portions of the etching barrier layer **134** on the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T that are exposed by the second photoresist pattern **142**. Also, the implanting angle of the second implant step **152** is different from the implanting angle of the first implant step **151**. According to some embodiments, the second implant step **152** is performed on the exposed portions of the etching barrier layer **134** at another angle which is different from the angle of the first implant step **151**, thereby introducing the dopants of a suitable conductivity type, such as P-type dopants (e.g. boron or another suitable doping material). As shown in FIG. 8A, the portions of the etching barrier layer **134** that are covered by the second photoresist pattern **142** are not implanted by any dopant. As shown in FIG. 8B, the positions of the implanted portions of the exposed etching barrier layer **134** correspond to the first sidewalls **115S1** and the top surfaces **115a** of the hard mask layers **115**.

(28) Next, after the second implant step **152** is completed, the second photoresist pattern **142** is removed, as shown in FIG. 9A and FIG. 9B. In some embodiments, the second photoresist pattern **142** is removed by an ashing process.

(29) As shown in FIG. 9A, the first implanted portions **134D1** of the etching barrier layer **134** are formed at the positions corresponding to the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T. The first implanted portions **134D1** of the etching barrier layer **134** are formed at the positions corresponding to the second sidewalls **115S2** of the hard mask layers **115**. Also, the first implanted portions **134D1** of the etching barrier layer **134** are formed on the portions of the first dielectric layer **132** that corresponds to the top surfaces **115a** of the hard mask layers **115**. The other portions of the etching barrier layer **134** that correspond to the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T are not implanted and can be referred as the first non-implanted portions **134U1**.

(30) As shown in FIG. 9B, the second implanted portions **134D2** of the etching barrier layer **134** are formed at the positions corresponding to the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T. The second implanted portions **134D2** of the etching barrier layer **134** are formed at the positions corresponding to the first sidewalls **115S1** of the hard mask layers **115**. Also, the second implanted portions **134D2** of the etching barrier layer **134** are formed on the portions of the first dielectric layer **132** that correspond to the top surfaces **115a** of the hard mask layers **115**. The other portions of the etching barrier layer **134** that correspond to the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T are not implanted and can be referred

as the second non-implanted portions **134U2**.

(31) Next, referring to FIG. **10A** and FIG. **10B**, the implanted portions of the etching barrier layer **134** are removed, while the non-implanted portions of the etching barrier layer **134** are remained. In some embodiments, the first implanted portions **134D1** and the second implanted portions **134D2** of the etching barrier layer **134** (such as a polysilicon layer) are removed by a wet etching process. The first non-implanted portions **134U1** (as shown in FIG. **10A**) and the second non-implanted portions **134U2** (as shown in FIG. **10B**) remain on the first dielectric layer **132**.

(32) As shown in FIG. **10A**, the remaining portions (i.e. the first non-implanted portion **134U1**) of the etching barrier layer **134** correspond to the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T. Also, the remaining portions (i.e. the first non-implanted portion **134U1**) of the etching barrier layer **134** expose the portions of the first dielectric layer **132** that correspond to the second sidewalls **115S2** and the top surfaces **115a** of the hard mask layers **115**. As shown in FIG. **10B**, the remaining portions (i.e. the second non-implanted portion **134U2**) of the etching barrier layer **134** correspond to the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T. Also, the remaining portions (i.e. the second non-implanted portion **134U2**) of the etching barrier layer **134** expose the portions of the first dielectric layer **132** that correspond to the first sidewalls **115S1** and the top surfaces **115a** of the hard mask layers **115**.

(33) Next, referring to FIG. **11A** and FIG. **11B**, parts of the first dielectric layer **132** that are uncovered by the non-implanted portions of the etching barrier layer **134** (i.e. the first non-implanted portion **134U1** and the second non-implanted portion **134U2**) are removed. In some embodiments, these parts of the first dielectric layer **132** are removed by wet etching and/or another suitable method (such as a SiCoNi etching method). Therefore, as shown in FIG. **11A**, the sidewall portions **132S1** of the first dielectric layer **132** are formed in the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T after the portions of the first dielectric layer **132** that correspond to the second sidewalls **115S2** and the top surfaces **115a** of the hard mask layers **115** are removed. In addition, as shown in FIG. **11B**, the sidewall portions **132S2** of the first dielectric layer **132** are formed in the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T after the portions of the first dielectric layer **132** that correspond to the first sidewalls **115S1** and the top surfaces **115a** of the hard mask layers **115** are removed.

(34) Next, referring to FIG. **12A** and FIG. **12B**, all of the portions of the etching barrier layer **134** remaining on the first dielectric layer **132** are removed. For example, the first non-implanted portion **134U1** in FIG. **11A** and the second non-implanted portion **134U2** in FIG. **11B** are removed. In some embodiments, the etching barrier layer **134** is a polysilicon layer. In one example, the etching barrier layer **134** is removed using an etching gas (such as sulfur hexafluoride; SF₆) with a high selectivity to polysilicon.

(35) As shown in FIG. **12A**, in the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T, the sidewall portions **132S1** of the first dielectric layer **132** remain on the first sidewalls **115S1** of the hard mask layers **115**, and the bottom portions **132B1** of the first dielectric layer **132** remain on the top surfaces **125a** of the bottom contact parts **125**. Also, as shown in FIG. **12B**, in the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T, the sidewall portions **132S2** of the first dielectric layer **132** remain on the second sidewalls **115S2** of the hard mask layers **115**, and the bottom portions **132B2** of the first dielectric layer **132** remain on the top surfaces **125a** of the bottom contact parts **125**.

(36) In the subsequent process, the portions of the first dielectric layer **132** in the bottoms of the first recesses **131** are removed to expose the top surfaces **125a** of the bottom contact parts **125**, in accordance with some embodiments. FIGS. **13A**, **13B**, **14A** and **14B** illustrate one of the methods for removing the portions of the first dielectric layer **132** in the bottoms of the first recesses **131**, in accordance with some embodiments of the present disclosure.

(37) Referring to FIG. **13A** and FIG. **13B**, a protective pattern **160** is formed on the hard mask layers **115**, thereby exposing the portions of the first dielectric layer **132** in the bottoms of the first

recesses **131**. In some embodiments, a polymer material layer or another suitable material layer is deposited on the structure as shown in FIG. **12A** and FIG. **12B** using methane (CH₄) as the deposition gas, followed by a suitable patterning process to form the protective pattern **160** as shown in FIG. **13A** and FIG. **13B**.

(38) As shown in FIG. **13A**, in the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T the protective pattern **160** covers the top surfaces **115a** of the hard mask layers **115**, the first nitride layers **121** on the second sidewalls **115S2** of the hard mask layers **115**, the first nitride layers **121** on the first sidewalls **115S1** of the hard mask layers **115**, and the sidewall portions **132S1** of the first dielectric layer **132** (on the first sidewalls **115S1** of the hard mask layers **115**).

(39) As shown in FIG. **13B**, in the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T, the protective pattern **160** covers the top surfaces **115a** of the hard mask layers **115**, the first nitride layers **121** on the first sidewalls **115S1** of the hard mask layers **115**, the first nitride layers **121** on the second sidewalls **115S2** of the hard mask layers **115**, and the sidewall portions **132S2** of the first dielectric layer **132** (on the second sidewalls **115S2**).

(40) Next, referring to FIG. **14A** and FIG. **14B**, the removal of the portions of the first dielectric layer **132** in the bottoms of the first recesses **131** is performed to expose the top surfaces **125a** of the bottom contact parts **125**. In some embodiments, the bottom portions **132B1** and **132B2** of the first dielectric layer **132** are removed by an etching process with a higher selectivity for the material of the first dielectric layer **132** than for the material of the protective pattern **160**.

(41) As shown in FIG. **14A**, the bottom portions **132B1** (FIG. **13A**) of the first dielectric layer **132** in the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T are removed, thereby exposing the top surfaces **125a**. In some embodiments, after the bottom portions **132B1** are removed, the top surfaces **122a** of the oxide layers **122** and the top surfaces **123a** of the second nitride layer **123** that are close to the second sidewalls **115S2** of the hard mask layers **115** are also exposed.

(42) Moreover, the bottom portions **132B2** (FIG. **13B**) of the first dielectric layer **132** in the first recesses **131** of the i-th row R.sub.iT and the (i-2)-th row R.sub.(i-2)T and the bottom portions **132B1** (FIG. **13A**) of the first dielectric layer **132** in the first recesses **131** of the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T can be removed simultaneously. As shown in FIG. **14B**, the bottom portions **132B2** (FIG. **13B**) are removed, thereby exposing the top surfaces **125a**. In some embodiments, after the bottom portions **132B2** of the first dielectric layer **132** are removed, the top surfaces **122a** of the oxide layers **122** and the top surfaces **123a** of the second nitride layer **123** that are close to the first sidewalls **115S1** of the hard mask layers **115** are also exposed.

(43) Next, after the top surfaces **125a** of the bottom contact parts **125** are exposed, the protective pattern **160** is removed. In some embodiments, the protective pattern **160** is removed by an ashing process, or another suitable process.

(44) The spacers with different thicknesses (such as the first spacers SP1 in FIG. **14A** and the second spacers SP2 in FIG. **14B**) can be formed through the above-mentioned processes as illustrated in FIGS. **5A-14B**, wherein the spacers are formed on the two opposite sidewalls of the hard mask layer **115** that protrude from the top surfaces **125a** of the bottom contact parts **125**. Also, the second recesses **161A** and **161B** are formed between the spacers. According to the embodiments, in the top view of the bottom contact parts **125** on the substrate **100**, the exposed top surfaces **125a** in the second recesses **161A** and the second recesses **161B** are misaligned in the second direction D2. It should be noted that the present disclosure is not limited to the processes as illustrated in FIGS. **5A-14B**. Other methods for forming spacers of different thicknesses on two opposite sidewalls of each of the hard mask layers **115** are also included in the present disclosure.

(45) As shown in FIG. **14A**, the first spacers SP1 are formed adjacent to the second recesses **161A** in the (i-1)-th row R.sub.(i-1)T and the (i-3)-th row R.sub.(i-3)T. Each of the first spacers SP1

includes the first nitride layer **121** on the first sidewall **115S1** of the hard mask layer **115** and the sidewall portion **132S1** of the first dielectric layer **132**. The first nitride layers **121** on the second sidewalls **115S2** of the hard mask layers **115** can be referred as the second spacers **SP2**. In the example shown in FIG. **14A**, the thickness of each of the first spacers **SP1** is greater than the thickness of each of the second spacers **SP2**.

(46) As shown in FIG. **14B**, the first spacers **SP1** are formed adjacent to the second recesses **161B** in the i -th row $R.sub.iT$ and the $(i-2)$ -th row $R.sub.(i-2)T$. The first nitride layers **121** on the first sidewalls **115S1** of the hard mask layers **115** can be referred as the first spacers **SP1**. Each of the second spacers **SP2** includes the first nitride layer **121** on the second sidewalls **115S2** of the hard mask layer **115** and the sidewall portion **132S1** of the first dielectric layer **132**. In the example shown in FIG. **14B**, the thickness of each of the first spacers **SP1** is less than the thickness of each of the second spacers **SP2**.

(47) Referring to FIGS. **15A**, **15B** and **15C**, several contacts $C.sub.A$ and $C.sub.B$ are formed on the substrate **100**. FIG. **15C** is a top view of the substrate **100**. FIGS. **15A** and **15B** are cross-sectional views taken along sectional lines A-A' and B-B' of the structure of FIG. **15C**. After the structures as shown in FIGS. **14A** and **14B** are formed, the top contact parts **165A** are formed in the second recesses **161A** (FIG. **15A**), and the top contact parts **165B** are formed in the second recesses **161B** (FIG. **15B**). As shown in FIG. **15A**, the top contact parts **165A** are formed on the bottom contact parts **125**. Each of the top contact parts **165A** and the underlying bottom contact part **125** are referred as a contact $C.sub.A$. As shown in FIG. **15B**, the top contact parts **165B** are formed on the bottom contact parts **125**. Each of the top contact parts **165B** and the underlying bottom contact part **125** are referred as a contact $C.sub.B$. As shown in FIG. **15C**, in the top view of the substrate **100**, those top contact parts **165A** and **165B** are arranged as the top contact parts **165B** of the i -th row $R.sub.iP$, the top contact parts **165A** of the $(i-1)$ -th row $R.sub.(i-1)P$, the top contact parts **165B** of the $(i-2)$ -th row $R.sub.(i-2)P$, and the top contact parts **165A** of the $(i-3)$ -th row $R.sub.(i-3)P$, wherein i is a positive integer. In addition, the top contact parts **165A** (or the contacts $C.sub.A$) and the top contact parts **165B** (or the contacts $C.sub.B$) in adjacent rows are misaligned with each other in the second direction **D2**.

(48) One of the methods for forming the top contact parts **165A** and **165B** is provided below. A second contact material layer is formed on the bottom contact parts **125**. The second contact material layer covers the top surfaces **115a** of the hard mask layers **115** and fills the second recesses **161A** and **161B** (FIG. **14A** and FIG. **14B**). The second contact material layer may include one or more metal materials, such as (but not limited to) tungsten. Next, the excess portions of the second contact material layer (i.e. the portions outside the second recesses **161A** and **161B**) are removed, and the remaining portions in the second recesses **161A** and **161B** form the top contact parts **165A** and **165B**, respectively.

(49) It should be noted that the top surfaces **165A-a** and **165B-a** of the top contact parts **165A** and **165B** are not higher than top surfaces **115a** of the hardmask layers **115** adjacent to the top contact parts **165A** and **165B**. That is, the contacts $C.sub.A$ and $C.sub.B$ are positioned between the hard mask layers **115** of the bit lines and not extended to the top surfaces **115a** of the hard mask layers **115**. In some embodiments, as shown in FIG. **15A**, the top surfaces **165A-a** are substantially level with the top surfaces **115a**. As shown in FIG. **15B**, the top surfaces **165B-a** are substantially level with the top surfaces **115a**.

(50) Next, referring to FIGS. **16A**, **16B** and **16C**, several contacts plugs **171** and **172** are formed on the contacts $C.sub.A$ and $C.sub.B$, respectively. FIG. **16C** is a top view of the substrate **100**. FIGS. **16A** and **16B** are cross-sectional views taken along sectional lines A-A' and B-B' of the structure of FIG. **16C**. After the structures as shown in FIGS. **15A** and **15B** are formed, a second dielectric layer **167** (such as a silicon nitride layer) is formed to cover the hard mask layers **115**, the first spacers **SP1**, the second spacers **SP2** and the contacts $C.sub.A$ and $C.sub.B$. Then, several contacts plugs **171** and **172** are formed on the contacts $C.sub.A$ and $C.sub.B$ and penetrate the second dielectric

layer **167**. As shown in FIG. **16A**, **16B**, the contact plugs **171**, **172** physically contact the contacts C.sub.A, C.sub.B respectively and electrically connect the contacts C.sub.A, C.sub.B respectively. Accordingly, the top surfaces **165A-a** of the top contact parts **165A** and the top surfaces **165B-a** of the top contact parts **165B** are landing surfaces of the contact plugs **171** and **172**, respectively. As shown in FIG. **16C**, the contact plugs **171** and **172** positioned in two adjacent rows are misaligned with each other in the second direction D2. Thus, the distance between the contact plugs **171** and **172** positioned in adjacent rows can be increased, thereby preventing electrical interference during operation.

(51) FIG. **17** is a cross-sectional view of a semiconductor structure, in accordance with some embodiments of the present disclosure. The components in FIG. **17** that are the same as those in the structure shown in FIG. **15B** use the same reference numerals. The materials and manufacturing methods of those components of the structure in FIG. **17** may be the same as, or similar to, those described in the embodiments above, and therefore the descriptions of these materials and manufacturing methods are not repeated herein.

(52) The semiconductor structure of the embodiment is, for example, applied to a dynamic random access memory (DRAM) device with buried word lines. Several active regions A.sub.A are formed in the substrate **100** of the memory device, and two adjacent active regions A.sub.A are separated by a trench isolation structure **102**. The memory device includes several buried word line (WL) sets **104** in the substrate **100**, and each of the buried WL sets **104** includes two buried word lines **104A** and **104B**. As shown in FIG. **17**, a buried word line (WL) sets **104** (including two separate buried word lines **104A** and **104B**) is formed in one of the active regions A.sub.A of the substrate **100**. In some embodiments, a bit line is formed above the adjacent buried word lines **104A** and **104B**, wherein the bit line includes the first conductive layer **113**, the second conductive layer **114** and the hard mask layer **115**. The buried word lines **104A** and **104B** are separated from the bit lines **116** by the insulation layers **105**. The contacts formed by the manufacturing method in accordance with the embodiments (such as the contacts C.sub.B in FIG. **17**) extend downwards to contact the surface of the substrate **100**.

(53) The aforementioned semiconductor structures and methods of manufacturing the semiconductor structures, in accordance with some embodiments of the present disclosure, have several advantages. For example, the contacts CA and CB are positioned between the hard mask layers **115** of the bit lines, and the top surfaces **165A-a** of the top contact parts **165A** and the top surfaces **165B-a** of the top contact parts **165B** are not higher than the top surfaces **115a** of the hard mask layers **115**, in accordance with some embodiments of the present disclosure. That is, the contacts (such as CA and CB) of the embodiments do not extend to the top surfaces **115a** of the hard mask layers **115**. The width W1 of the top surface and the width W2 of the bottom surface of the top contact part **165A**, and the width W3 of the top surface of the bottom contact part **125** are shown in FIG. **17**. In the existing method for manufacturing the semiconductor structure, especially in the manufacturing process of small-size memory devices, a complicated and expensive process is required to complete the formation of the contacts extending to the top surfaces of the bit lines and the contact plugs subsequently connected to these contacts. Compared with the existing method for manufacturing the semiconductor structure, the semiconductor structure provided in the embodiments can be formed using a general photoresist with a typical photolithography process (for example, the photoresist patterns with the photoresist pattern strips **1411**, **1412**, **1421** and **1422** as shown in FIG. **5C** and FIG. **7C** having the width greater than the width of the recess) to form the contacts (such as CA and CB) of the embodiments. Therefore, the semiconductor structure and the method of manufacturing the same as provided in accordance with some embodiments do decrease the process complexity and save the production cost. Furthermore, the semiconductor structure as manufactured in accordance with some embodiments has components with complete profiles (for example, the top surfaces of the contacts having sufficient widths can be the landing surfaces for the contact plugs), and thus has excellent electrical characteristics. Therefore, the memory device

applied with the semiconductor structure of the embodiment has good electrical reliability and stable operation performance.

(54) While the disclosure has been described by way of example and in terms of the preferred embodiments, it should be understood that the disclosure is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. A semiconductor structure, comprising: a substrate; bit lines on the substrate, wherein each of the bit lines comprises: a first conductive layer on the substrate; a second conductive layer on the first conductive layer; and a hardmask layer on the second conductive layer; contacts disposed on the substrate and positioned between two adjacent bit lines, wherein bottom surfaces of the contacts physically contact the substrate, and top surfaces of the contacts are not higher than top surfaces of the hardmask layers adjacent to the contacts, wherein each of the contacts comprises a bottom contact part on the substrate and a top contact part on the bottom contact part, and a width of a top surface of the top contact part is greater than a width of a top surface of the bottom contact part, wherein the top contact part covers a first lower spacer portion and does not cover a second lower spacer portion that are respectively formed on opposite sidewalls of the bottom contact part; and a plurality of contact plugs respectively disposed on the contacts and partially extending into the contacts, wherein a bottommost portion of each of the contact plugs is disposed within the corresponding contact, and a sidewall of the bottommost portion is in direct contact with a sidewall of the corresponding contact.
2. The semiconductor structure as claimed in claim 1, wherein the top contact part is positioned between two adjacent hardmask layers, wherein the top surfaces of the top contact parts are level with the top surfaces of the hardmask layers.
3. The semiconductor structure as claimed in claim 2, wherein two opposite sidewalls of each of the top contact parts respectively abut spacers on two adjacent hardmask layers, and the spacers have different thicknesses.
4. The semiconductor structure as claimed in claim 1, wherein each of the hardmask layers has a first sidewall and a second sidewall opposite to the first sidewall, and the semiconductor structure further comprises: first spacers on the first sidewalls; and second spacers on the second sidewalls, wherein a thickness of each of the first spacers is different from a thickness of each of the second spacers.
5. The semiconductor structure as claimed in claim 4, wherein each of the first spacers comprises a first nitride layer, and each of the second spacers comprises another first nitride layer and a second nitride layer.
6. The semiconductor structure as claimed in claim 4, wherein each of the first spacers comprises a first nitride layer and a second nitride layer, and each of the second spacers comprises the first nitride layer.
7. The semiconductor structure as claimed in claim 1, wherein the top surfaces of the top contact parts are not higher than the top surfaces of the hardmask layers adjacent to the contacts.
8. The semiconductor structure as claimed in claim 1, wherein the top contact part of each of the contacts is disposed on one of the first lower spacer portion and the second lower spacer portion, and the one of the first lower spacer portion and the second lower spacer portion is disposed between the bottom contact part and the bit line disposed adjacent to the one of the first lower spacer portion and the second lower spacer portion, wherein a bottom surface of the top contact part covers the top surface of the underlying bottom contact part and a top surface of the underlying

one of the first lower spacer portion and second lower spacer portion.

9. The semiconductor structure as claimed in claim 1, wherein the top surfaces of the bottom contact parts of the contacts are higher than top surfaces of the second conductive layers of the bit lines.

10. The semiconductor structure as claimed in claim 1, wherein the contacts are arranged along a first direction to form a plurality of rows, and the contacts of two adjacent rows are misaligned with each other in a second direction, wherein the second direction is different from the first direction.

11. The semiconductor structure as claimed in claim 10, wherein the contact plugs of two adjacent rows are misaligned with each other in the second direction.

12. A method of manufacturing the semiconductor structure of claim 1, comprising: providing a substrate and bit lines formed on the substrate, and a spacer material layer covering sidewalls of a top surface of each of the bit lines, wherein each of the bit lines comprises a conductive layer and a hardmask layer on the conductive layer; forming a first contact material layer on the substrate, the first contact material layer covering the spacer material layers on the bit lines; etching the first contact material layer to remove a portion of the first contact material layer, a portion of the hardmask layers and a portion of the spacer material layers, thereby recessing the first contact material layer and forming first recesses between remaining portions of the hardmask layers, wherein remaining portions of the first contact material layer form bottom contact parts; depositing a first dielectric layer over the bottom contact parts, the first dielectric layer conformally formed on sidewalls and bottom surfaces of the first recesses and top surfaces of the remaining portions of the hardmask layers; patterning the first dielectric layer to remove a portion of the first dielectric layer and expose top surfaces of the bottom contact parts, thereby forming spacers with different thicknesses on two opposite sidewalls of the hardmask layers that protrude from the bottom contact parts, wherein second recesses are formed between two adjacent spacers, and the second recesses expose the top surfaces of the bottom contact parts; forming top contact parts on the bottom contact parts, wherein top surfaces of the top contact parts are not higher than the top surfaces of the hardmask layers adjacent to the top contact parts, and each of the top contact parts covers a first lower spacer portion and does not cover a second lower spacer portion that are respectively formed on opposite sidewalls of the underlying bottom contact part; and forming a plurality of contact plugs respectively disposed on the top contact parts and partially extending into the top contact parts, wherein a bottommost portion of each of the contact plugs is disposed within the corresponding contact, and a sidewall of the bottommost portion is in direct contact with a sidewall of the corresponding contact.

13. The method of manufacturing the semiconductor structure as claimed in claim 12, wherein the top surfaces of the top contact parts are level with the top surfaces of the hardmask layers.

14. The method of manufacturing the semiconductor structure as claimed in claim 12, wherein the hardmask layers that protrude from the bottom contact parts have first sidewalls and second sidewalls opposite to the first sidewalls, first spacers are formed on the first sidewalls and second spacers are formed on the second sidewalls, wherein thicknesses of the first spacers are different from thicknesses of the second spacers.

15. The method of manufacturing the semiconductor structure as claimed in claim 14, wherein the first recesses are arranged along a first direction to form a plurality of rows, and the first spacers on the first sidewalls of the hardmask layers in different rows have different thicknesses.

16. The method of manufacturing the semiconductor structure as claimed in claim 12, wherein the first recesses are arranged along a first direction to form a plurality of rows, and patterning the first dielectric layer comprises: performing different patterning steps on portions of the first dielectric layer that are conformally formed on the first recesses arranged in adjacent rows.

17. The method of manufacturing the semiconductor structure as claimed in claim 16, wherein patterning the first dielectric layer comprises: conformally depositing an etching barrier layer on the first dielectric layer; providing a first photoresist pattern on the first recesses of i-th row,

wherein i is a positive integer, and the first photoresist pattern exposes the etching barrier layer on the first recesses of the $(i-1)$ -th row; performing a first implant step on portions of the etching barrier layer on the first recesses of the $(i-1)$ -th row to form first implanted portions, wherein the first implanted portions are formed on respective portions of the first dielectric layer that are formed on second sidewalls and the top surfaces of the hardmask layers; and removing the first photoresist pattern.

18. The method of manufacturing the semiconductor structure as claimed in claim 17, wherein patterning the first dielectric layer further comprises: providing a second photoresist pattern on the first recesses of the $(i-1)$ -th row, wherein the second photoresist pattern exposes the etching barrier layer on the first recesses of the i -th row; performing a second implant step on portions of the etching barrier layer on the first recesses of the i -th row to form second implanted portions, wherein the second implanted portions are formed on other respective portions of the first dielectric layer that are formed on first sidewalls and the top surfaces of the hardmask layers, wherein the first sidewalls are opposite to the second sidewalls; and removing the second photoresist pattern.

19. The method of manufacturing the semiconductor structure as claimed in claim 18, wherein patterning the first dielectric layer further comprises: removing the first implanted portions and the second implanted portions, thereby leaving non-implanted portions of the etching barrier layer; removing parts of the first dielectric layer that are uncovered by the non-implanted portions of the etching barrier layer; removing the non-implanted portions of the etching barrier layer; and removing other parts of the first dielectric layer that are disposed in bottom surfaces of the first recesses, thereby exposing the top surfaces of the bottom contact parts.

20. The method of manufacturing the semiconductor structure as claimed in claim 19, wherein removing parts of the first dielectric layer that are uncovered by the non-implanted portions of the etching barrier layer comprises: removing first parts of the first dielectric layer that are formed on the second sidewalls of the hardmask layers at the first recesses of the $(i-1)$ -th row; and removing second parts of the first dielectric layer that are formed on the first sidewalls of the hardmask layers at the first recesses of the i -th row.
